

Hardware Project No. 1

Instrumentation Amplifier

ECE 444: Bioinstrumentation and Sensing

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May 12, 2026

1. Introduction

Instrumentation amplifiers are the workhorse front-end circuits in bioinstrumentation and sensing applications. Biopotential signals such as electrocardiograms (EKG), electromyograms (EMG), and electroencephalograms (EEG) are small differential voltages, typically in the microvolt to millivolt range, that appear in the presence of much larger common-mode interference (60 Hz line pickup being the most familiar example). A useful measurement amplifier must therefore provide high differential gain, very high input impedance so as not to load the signal source, and strong rejection of common-mode signals.

This report describes the design, construction, and characterization of a three-op-amp instrumentation amplifier built from discrete LF411 op amps and 5% resistors. The assignment required a flat frequency response from 5 Hz to 1 kHz and a minimum differential gain of 200. The circuit was designed for a nominal gain of approximately 500, which provides ample margin above the specified minimum while keeping the output amplitude conveniently measurable on the oscilloscope without saturating the op-amp output stages. In addition to the frequency response, the circuit was characterized for common-mode rejection ratio (CMRR) and quiescent power consumption.

2. Theory and Design Derivation

2.1 Three-Op-Amp Architecture

The standard three-op-amp instrumentation amplifier consists of two cascaded stages. The input stage is a pair of non-inverting amplifiers sharing a common gain-setting resistor R_1 between their inverting input nodes, with feedback resistors R_2 from each op-amp output back to its inverting input. The output stage is a unity-gain difference amplifier built from a single op amp and four equal resistors R . The topology is shown in Figure 1.

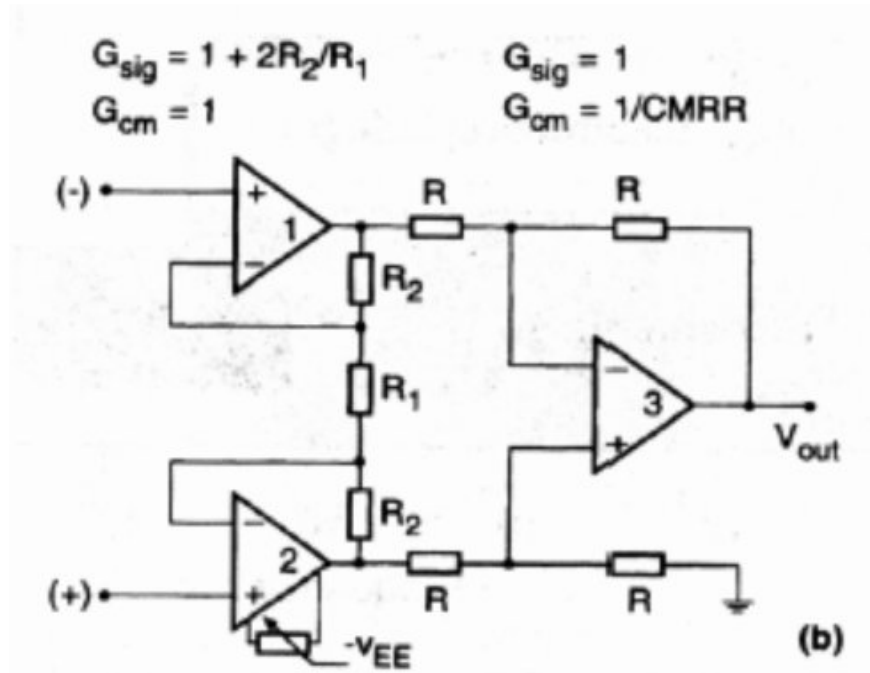


Figure 1. Three-op-amp instrumentation amplifier topology, reproduced from the assignment handout (originally from *The Biomedical Engineering Handbook*). The amplifier's negative input (-) connects to A1's non-inverting terminal and the positive input (+) connects to A2's non-inverting terminal. As built: $R_1 = 400\ \Omega$ (two $200\ \Omega$ resistors in series), $R_2 = 100\ \text{k}\Omega$ (each of the two first-stage feedback resistors), $R = 10\ \text{k}\Omega$ (each of the four difference-amplifier resistors). All three op amps are LF411, powered from $\pm 12\ \text{V}$ supplies.

2.2 First-Stage Differential Gain

The first-stage gain is derived using only the two ideal op-amp rules: (i) the two input terminals of an op amp are at the same voltage under negative feedback (the virtual-short condition), and (ii) no current flows into the op-amp input terminals. Following the labeling in Figure 1, let V_- denote the voltage at A1's non-inverting input (the amplifier's negative input terminal) and V_+ denote the voltage at A2's non-inverting input (the amplifier's positive input terminal). Let V_{o1} and V_{o2} denote the corresponding op-amp output voltages.

By the virtual-short rule, the inverting input of A1 sits at V_- and the inverting input of A2 sits at V_+ . The two inverting inputs are connected together through R_1 alone (the same nodes also drive the two R_2 feedback paths, but no current flows into the op-amp inputs themselves). The current through R_1 , taken as positive when flowing from A1's inverting input toward A2's inverting input, is therefore

$$I_{R1} = (V_- - V_+) / R_1.$$

Because no current flows into either op-amp input, this same current I_{R1} flows from V_{o1} , through the upper R_2 , into A1's inverting input, then through R_1 to A2's inverting input, then through the lower R_2 to V_{o2} . Applying Ohm's law to the upper R_2 :

$$\begin{aligned} V_{o1} - V_- &= I_{R1} \cdot R_2 = (V_- - V_+) \cdot R_2 / R_1, \\ V_{o1} &= V_- + (V_- - V_+) \cdot R_2 / R_1. \end{aligned}$$

Applying Ohm's law to the lower R_2 , where the same current flows out of A2's inverting input toward V_{o2} :

$$V_+ - V_{o2} = I_{R1} \cdot R2 = (V_- - V_+) \cdot R2 / R1,$$

$$V_{o2} = V_+ - (V_- - V_+) \cdot R2 / R1.$$

The differential output of the input stage is therefore

$$V_{o1} - V_{o2} = (V_- - V_+) \cdot (1 + 2 R2 / R1).$$

Equivalently, taking the opposite sign convention (which the difference-amplifier stage will impose in Section 2.3),

$$V_{o2} - V_{o1} = (V_+ - V_-) \cdot (1 + 2 R2 / R1),$$

so the first-stage differential gain, defined with respect to $(V_+ - V_-)$, is

$$G_{sig} = 1 + 2 R2 / R1.$$

If instead a common-mode voltage V_{cm} is applied to both inputs ($V_+ = V_- = V_{cm}$), then $I_{R1} = 0$ and no current flows through either $R2$. Both op-amp outputs sit at V_{cm} . The input-stage common-mode gain is therefore exactly unity, which is the key property of this stage: it amplifies the differential signal by G_{sig} while passing the common-mode signal through unchanged, raising the differential-to-common-mode amplitude ratio at this internal node by exactly G_{sig} relative to the input.

2.3 Second-Stage Difference Amplifier

The output stage is a standard difference amplifier with all four resistors equal to R . Following the wiring in Figure 1, V_{o1} (the output of A1) drives A3's inverting input through a series R , with a feedback R from the output V_{out} back to that same inverting node. V_{o2} (the output of A2) drives A3's non-inverting input through a series R , with another R from the non-inverting node to ground. Applying the same two ideal op-amp rules:

At the non-inverting node of A3, the voltage divider from V_{o2} to ground gives

$$V_+(A3) = V_{o2} \cdot R / (R + R) = V_{o2} / 2.$$

By the virtual-short rule, the inverting input of A3 also sits at $V_{o2}/2$. Applying KCL at the inverting node, with no current flowing into the op-amp input:

$$(V_{o1} - V_{o2}/2) / R = (V_{o2}/2 - V_{out}) / R,$$

$$V_{o1} - V_{o2}/2 = V_{o2}/2 - V_{out},$$

$$V_{out} = V_{o2} - V_{o1}.$$

The second-stage differential gain is therefore unity, with the sign convention that $V_{out} = V_{o2} - V_{o1}$. Combining this with the first-stage result $V_{o2} - V_{o1} = (V_+ - V_-)(1 + 2 R2 / R1)$ from Section 2.2, the overall amplifier output is

$$V_{out} = (V_+ - V_-) \cdot (1 + 2 R2 / R1).$$

Importantly, the difference-amplifier stage takes the difference $V_{o2} - V_{o1}$, so any common-mode component that survives the first stage (which is the entire common-mode input, since G_{cm} of the input stage is unity) is cancelled to the extent that the four R resistors are exactly matched. Mismatch in these four R values directly converts a fraction of the common-mode input into differential output, which is the dominant CMRR limit in real circuits built from finite-tolerance resistors. This point is quantified for the measured CMRR in Section 6.3.

3. Component Selection

The design target was a nominal gain near 500. Solving $1 + 2 R_2/R_1 = 501$ with the convenient choice $R_2 = 100 \text{ k}\Omega$ gives $R_1 = 400 \text{ }\Omega$. This R_1 was implemented as two $200 \text{ }\Omega$ resistors in series because no single $400 \text{ }\Omega$ resistor of suitable tolerance was on hand. The difference-stage resistors were chosen as $R = 10 \text{ k}\Omega$, a moderate value large enough to keep the op-amp output currents low while small enough that op-amp input bias currents do not produce significant offset on the high-impedance nodes. The selected nominal gain is

$$A_{d,nominal} = 1 + 2(100 \text{ k}\Omega)/(400 \text{ }\Omega) = 501, \text{ or } 20 \log_{10}(501) = 54.00 \text{ dB.}$$

Component	Nominal Value	Function
R1	400 Ω (two 200 Ω in series)	Gain-setting resistor (sets G_{sig})
R2 ($\times 2$)	100 k Ω	First-stage feedback resistors
R ($\times 4$)	10 k Ω	Difference-amplifier resistors
A1, A2, A3	LF411 (JFET-input op amps)	Three active devices
Supplies	$\pm 12 \text{ V}$ dual rail	Bench DC supply, dual rail

Table 1. Component values used in the instrumentation amplifier. Individual resistors were not hand-measured; reported values are nominal at 5% tolerance.

The LF411 was selected as the active element. It is a single JFET-input op amp with typical gain-bandwidth product of approximately 3 MHz, slew rate of approximately 13 V/ μ s, and very low input bias current. The JFET input stage is useful here because the amplifier uses relatively large feedback resistances, especially the 100 k Ω resistors in the first stage. Low input bias current reduces DC errors caused by bias currents flowing through these resistors. In this circuit, the measured frequency response is also consistent with the LF411 gain-bandwidth limitation at high closed-loop gain.

4. Experimental Procedure

The circuit was constructed on a solderless breadboard. For each LF411, pin 7 was connected to the +12 V rail, pin 4 to the -12 V rail, pin 3 to the non-inverting input node, pin 2 to the inverting input and feedback network, and pin 6 to the output.

For the frequency-response test, the amplifier was driven in a pseudo-differential configuration: the negative input terminal was tied to circuit ground, and the positive input terminal was driven by a sinusoidal signal of 5 mV_{pp}. Because the differential input is defined as $V_+ - V_-$, this is equivalent to applying a 5 mV_{pp} differential input. With the nominal gain of 501, the expected output amplitude is

$$V_{out,pp,expected} = 501 \cdot 5 \text{ mV}_{pp} = 2.505 \text{ V}_{pp},$$

which is large enough to read precisely on the oscilloscope while remaining well below the $\pm 10 \text{ V}$ output swing limit of the LF411. Frequency-response measurements were taken at 13 points from 5 Hz to 5 kHz, with denser spacing at low frequencies as suggested in the assignment (5, 10, 20, 50 Hz) and then at 100, 200, 300, 500, 700, 1000, 1500, 2000, and 5000 Hz to capture the onset of high-frequency rolloff.

For the CMRR test, the two amplifier inputs were tied together and driven by a single 100 Hz sinusoid. Under this condition the differential input is identically zero by construction, and any non-zero output is due entirely to the finite common-mode gain of the realized circuit. The common-mode input amplitude was 1.00 V_{pp}, which is large enough to produce a measurable common-mode output.

For the power-consumption measurement, the supply voltages and currents drawn from each rail were read directly from the bench supply digital display, and the total quiescent power was computed using the magnitudes of the positive and negative supply rails:

$$P = V_+ \cdot I_+ + |V_-| \cdot I_-.$$

Representative oscilloscope screenshots were recorded at selected frequencies rather than at every measurement point. This follows the assignment guidance to provide representative printouts rather than all individual frequency measurements. A photograph of the completed breadboard layout was also included to document the physical circuit construction.

5. Results

5.1 Frequency Response Data

Table 2 presents the measured frequency response. The gain at each frequency was computed as $G = V_{out,pp} / V_{in,pp}$ and converted to decibels as $20 \log_{10}(G)$.

Freq (Hz)	Vout (Vpp)	Vin (Vpp)	Gain (V/V)	Gain (dB)
5	2.50	0.005	500	53.98
10	2.57	0.005	514	54.22
20	2.57	0.005	514	54.22
50	2.59	0.005	518	54.29
100	2.60	0.005	520	54.32
200	2.57	0.005	514	54.22
300	2.57	0.005	514	54.22
500	2.55	0.005	510	54.15
700	2.55	0.005	510	54.15
1000	2.53	0.005	506	54.08
1500	2.49	0.005	498	53.94
2000	2.43	0.005	486	53.73
5000	2.00	0.005	400	52.04

Table 2. Measured frequency response. Input amplitude held at 5 mVpp throughout.

5.2 Bode Plot

Figure 2 plots the measured gain in decibels on a semilogarithmic frequency axis, overlaid with a single-pole low-pass fit $V_{out}/V_{in} = K/(1 + j f/f_c)$ with $K = 520$ and $f_c = 6.02$ kHz.

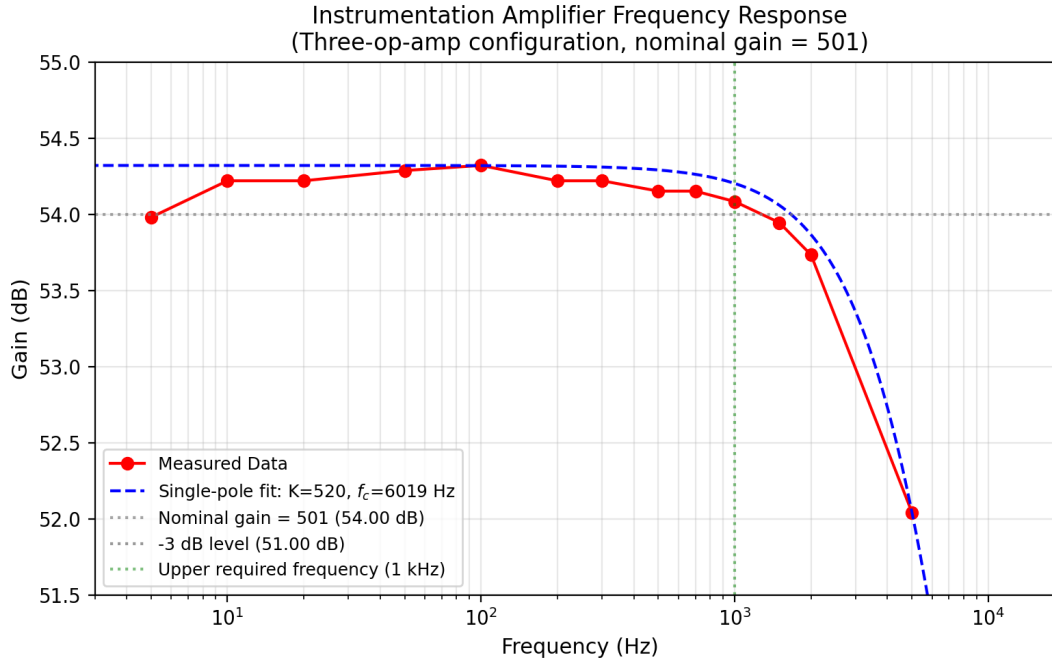


Figure 2. Measured instrumentation amplifier frequency response (red), overlaid with a single-pole rolloff fit (blue dashed) using $K = 520$ and $f_c = 6019$ Hz. The fit corner is set by extrapolating the measured 5 kHz attenuation back through a one-pole transfer function; see Section 6.2.

5.3 Key Performance Metrics

Metric	Value
Nominal differential gain	$A_d = 1 + 2R_2/R_1 = 501$ (54.00 dB)
Measured gain at 100 Hz	$A_{d,meas} = 520$ (54.32 dB)
Gain error vs. nominal	+3.8% (well within 5% resistor tolerance budget)
Gain range, 5 Hz to 1 kHz	500 to 520 (54.0 to 54.3 dB)
Gain at 5 kHz	400 (52.04 dB), -2.28 dB from 100 Hz value
Single-pole fit corner frequency	$f_c \approx 6.02$ kHz
LF411 GBW (typ.)	3 MHz
First-stage noise gain	$1 + 2R_2/R_1 = 501$
Predicted closed-loop -3 dB	$GBW / \text{noise gain} \approx 5.99$ kHz

Table 3. Summary of frequency-response metrics with theoretical comparisons.

5.4 Common-Mode Rejection Ratio

With both inputs tied together and driven by a 100 Hz, 1.00 V_{pp} sinusoid, the measured common-mode output was

$$V_{out,cm,pp} = 80 \text{ mV}_{pp} = 0.080 \text{ V}_{pp}.$$

The common-mode gain at 100 Hz is therefore

$$A_{cm} = V_{out,cm,pp} / V_{cm,pp} = 0.080 / 1.00 = 0.080.$$

Using the measured differential gain $A_d = 520$ at the same frequency, the common-mode rejection ratio is

$$CMRR = A_d / A_{cm} = 520 / 0.080 = 6500, \quad CMRR_{dB} = 20 \log_{10}(6500) = 76.3 \text{ dB}.$$

5.5 Power Consumption

Supply currents were read directly from the bench supply digital display:

Rail	Measured Voltage	Measured Current	Power
+ rail	11.999 V	0.008 A	0.0960 W
– rail (magnitude)	12.010 V	0.008 A	0.0961 W
Total	—	—	0.192 W = 192 mW

Table 4. Quiescent power consumption from the bench supply display.

The total measured supply current was approximately 16 mA, with 8 mA drawn from each supply rail. This corresponds to a total quiescent power consumption of approximately 192 mW. This current draw is reasonable for a circuit using three LF411 op amps powered from ± 12 V supplies. The measured current includes the quiescent current of all three op amps as well as any additional current drawn through the external resistor network and breadboard wiring.

6. Discussion

6.1 Gain Accuracy

The measured gain at 100 Hz of 520 V/V is 3.8% above the nominal value of 501. This is well within the tolerance budget set by the 5% resistors used to build the circuit. The gain is strongly dependent on the ratio R_2/R_1 . In the high-gain limit, fractional error in R_1 appears almost directly as an opposite fractional error in gain, and fractional error in the effective R_2 value appears almost directly as a same-sign fractional error in gain. Therefore, a measured gain error of 3.8% is fully consistent with the use of 5% resistors. Since R_1 was built from two nominal 200 Ω resistors in series and the R_2 resistors were nominal 100 k Ω , ordinary resistor tolerance alone is enough to explain the difference between the nominal gain of 501 and the measured gain of 520. No additional circuit failure or unexpected effect is needed to explain the gain error.

6.2 Frequency Response and Bandwidth Limitation

The amplifier is essentially flat from 5 Hz through approximately 1 kHz, varying only over the range 500 to 520 V/V (less than 0.35 dB peak-to-peak across the specified passband). Above 1 kHz the gain begins to roll off in a manner consistent with a single-pole low-pass response. Fitting the data to $V_{out}/V_{in} = K / (1 + j f/f_c)$ using the 5 kHz attenuation (gain dropped to 400 V/V from a passband value of 520 V/V) yields

$$f_c = 5000 \text{ Hz} / \sqrt{[(520/400)^2 - 1]} \approx 6.02 \text{ kHz}.$$

This corner frequency is set by the gain-bandwidth product of the LF411. For each input-stage op amp, the noise gain (the inverse of the feedback factor β) is equal to the full differential gain of the first stage, namely $1 + 2 R_2/R_1 = 501$. The reason is that, for differential operation, the midpoint of R_1 is a virtual ground by symmetry, so each input op amp sees a feedback divider of $R_1/2$ to virtual ground and R_2 to its output, giving $1/\beta = 1 + R_2/(R_1/2) = 1 + 2 R_2/R_1 = 501$. The expected closed-loop –3 dB bandwidth is therefore

$$f_{-3dB} \approx GBW / \text{noise gain} = 3 \text{ MHz} / 501 \approx 5.99 \text{ kHz}.$$

This agreement suggests that the observed high-frequency rolloff is primarily gain-bandwidth limited. The measured response does not prove a perfect single-pole model, since only a small number of data points were taken above 1 kHz, but the fitted corner frequency of approximately 6.02 kHz agrees closely with the expected $3 \text{ MHz} / 501 \approx 5.99 \text{ kHz}$ bandwidth estimate. Therefore, the LF411 gain-bandwidth product provides a plausible explanation for the observed decrease in gain above the required 1 kHz operating range. The second stage (A3) has unity differential gain, so its closed-loop bandwidth is the full GBW of A3, around 3 MHz, far beyond the input-stage limit, and does not contribute to the observed rolloff.

6.3 Common-Mode Rejection

The measured CMRR of 76.3 dB at 100 Hz is reasonable for a breadboarded instrumentation amplifier built from discrete 5% resistors. The first stage increases the differential signal by approximately 501 while passing common-mode signals with approximately unity gain, so the final CMRR is strongly affected by how well the four 10 k Ω resistors in the difference amplifier match each other.

The practical implication of the measured CMRR is that a 1.00 Vpp common-mode input signal produced a residual output of 80 mVpp, not zero. This corresponds to a common-mode gain of $A_{cm} = 0.080$. However, when this residual output is referred back to the amplifier input by dividing by the measured differential gain $A_d = 520$, it is equivalent to a differential input error of $80 \text{ mVpp} / 520 \approx 0.154 \text{ mVpp}$. Thus, the amplifier substantially rejects common-mode interference, but the remaining common-mode output is still large enough to matter for very small biopotential signals. The measured rejection is mainly limited by resistor mismatch in the final difference-amplifier stage rather than by the intrinsic CMRR of the LF411 op amps.

6.4 Construction and Troubleshooting

During initial testing the amplifier produced a rail-to-rail square-wave output rather than an amplified sinusoid, with the output sitting near the $\pm 12 \text{ V}$ supply rails. This is the unmistakable signature of either saturation or positive feedback. The error was traced to incorrect wiring of the A3 difference-amplifier stage: a transposed pair of input wires had connected the wrong op-amp outputs to A3's inverting and non-inverting inputs, which (combined with the feedback R) converted the difference amplifier into a positive-feedback configuration that latched to whichever rail it powered up against. Correcting the A3 wiring produced the expected linear amplification with a 5 mVpp input giving roughly 2.5 Vpp output, consistent with a gain near 500. This experience reinforced the importance of careful schematic-to-breadboard verification, particularly for the difference-amplifier stage where the input polarity is sensitive.

6.5 Error Sources

Listed in order of contribution to observed deviations from ideal:

1. Resistor tolerance (dominant for gain). The 3.8% gain error is consistent with 5% tolerance on the R1 (gain-setting) resistors, which is the dominant single source of error in the gain magnitude itself.
2. Difference-amplifier resistor mismatch (dominant for CMRR). Mismatch among the four $R = 10 \text{ k}\Omega$ resistors in the difference amplifier directly converts a fraction of the common-mode input into differential output, and is the dominant limit on the measured CMRR. With 5% resistors, the practical CMRR achievable from this stage is bounded; using precision-matched resistors or a monolithic in-amp would substantially improve it.

3. LF411 gain-bandwidth product (dominant for upper bandwidth). The first-stage noise gain of 501 against the LF411's 3 MHz typical GBW sets the closed-loop -3 dB bandwidth at approximately 6 kHz, comfortably above the required 1 kHz upper limit.

4. Oscilloscope and function-generator measurement uncertainty. Reading peak-to-peak amplitudes to three significant figures on a digital scope introduces approximately $\pm 2\%$ uncertainty in each gain measurement, which contributes to the small flutter visible in the 5 Hz to 200 Hz passband data (gain values jumping between 500 and 520 V/V).

5. Breadboard parasitics. Stray capacitance (typically 2–5 pF between adjacent rows) and contact resistance are small contributors at frequencies below the LF411's GBW-limited bandwidth, and at the relatively low node impedances of this circuit (10 k Ω to 100 k Ω), they do not significantly degrade performance in the 5 Hz to 1 kHz passband.

6. Op-amp non-idealities (DC offset, input bias current). The LF411's JFET input stage gives a typical input bias current of 50 pA, which produces approximately 0.5 μ V of DC offset at the difference-amplifier node, far below the millivolt-scale signals being amplified.

6.6 Improvements

For a more demanding implementation, several improvements would be straightforward. Using 0.1% precision matched resistors for the four difference-amplifier resistors would substantially improve the CMRR contribution from the difference-amplifier stage. Replacing the discrete three-op-amp implementation with a commercial monolithic instrumentation amplifier (INA128, INA333, or AD620) would provide laser-trimmed internal resistors with typical CMRR of 100–120 dB, while also reducing parts count and simplifying the gain-setting to a single external resistor. A printed-circuit-board implementation would reduce stray capacitance and contact resistance compared to the breadboard, although these are not currently limiting performance.

7. Conclusion

A three-op-amp instrumentation amplifier was designed, built, and characterized. The first-stage differential gain was derived from the op-amp ideal rules to be $G_{sig} = 1 + 2 R_2/R_1$, and the second-stage difference amplifier was shown to have unity differential gain, giving an overall differential gain of $1 + 2 R_2/R_1$. With $R_1 = 400 \Omega$ (two 200 Ω resistors in series), $R_2 = 100 \text{ k}\Omega$, and $R = 10 \text{ k}\Omega$ in the difference amplifier, the nominal gain is 501 (54.00 dB).

Experimentally, the measured gain at 100 Hz was 520 (54.32 dB), in agreement with nominal to 3.8% and consistent with ordinary 5% resistor tolerance on R_1 . The gain remained between 500 and 520 V/V across the specified 5 Hz to 1 kHz passband, satisfying both the bandwidth requirement and the minimum-gain requirement of 200. Above 1 kHz the response rolled off in a manner consistent with a single-pole low-pass behavior at approximately 6 kHz, which agrees closely with the LF411's typical 3 MHz gain-bandwidth product divided by the first-stage noise gain of 501.

The measured CMRR at 100 Hz was 76.3 dB, which is a reasonable value for a breadboarded instrumentation amplifier built from discrete 5% resistors and is limited primarily by the resistor matching in the final difference-amplifier stage rather than by the intrinsic CMRR of the LF411 op amps. Total quiescent power consumption from the ± 12 V supplies was 192 mW. Overall, the circuit met all specifications of the assignment.

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